OligoTox-CNS

An open, sequence-resolved dataset of central-nervous-system toxicity for oligonucleotide therapeutics

NIH/NCATS Oligonucleotide Toxicity Open Data Challenge — Phase 2, Data Generation · Narrative document · release v1.0

Oligonucleotides	CNS measurements	Modification records	Sources	Structural QC
1,839	**2,065**	**32,569**	**5**	**30/30 pass**

1 Executive summary

OligoTox-CNS pairs the **design** of 1,839 oligonucleotides — their sequences and the position of every chemical modification within them — with 2,065 measured central-nervous-system toxicity outcomes. 1,830 of the 1,839 compounds (99.5%) carry a full per-nucleotide chemistry map, expanded into a 32,569-row position table. This is the pairing a predictive model needs, and it is the thing the public literature has not previously offered in one structured place for the CNS.

The scientific core is the supplementary data of Hagedorn et al. (2022), released under CC BY. That table is widely cited for its 148 mouse-dosed compounds; it in fact contains 1,825 oligonucleotides, each with a sequence whose upper/lower case encodes locked-nucleic-acid position, and each with a measured rat primary-neuron calcium-oscillation score. 181 of them additionally carry a mouse acute tolerability score. Restructuring the full table — rather than the 148 — and adding a verified per-position modification map is this release's principal contribution.

Three further sources widen the picture: a divalent-cation formulation-rescue experiment in mice, a late-onset neurotoxicity study whose per-position chemistry had to be recovered from PDF typeface, and the FDA prescribing information for the two approved intrathecal antisense drugs. A fifth source contributes measurement instruments but no rows.

Positive and negative controls

The dataset is not a list of toxic compounds; it spans the full severity range and contains deliberate controls at both ends, which is what makes it trainable.

- **Designed negative controls (13).** Guanine-free antisense oligonucleotides with no perfectly or partially matching target in the mouse transcriptome, built by the source authors as a negative class. Their mouse tolerability scores run from 0 to 0.67 on a 0–20 scale.
- **Vehicle control.** Phosphate-buffered saline, scored on the same scale (1.67), which fixes the assay's floor: a score of roughly 1–2 is what an untreated animal produces.
- **A non-toxic clinical-chemistry control.** A 5-10-5 2′-MOE gapmer already used in clinical trials, included by its source authors specifically as a compound that does *not* produce the late-onset phenotype.
- **Severe positive controls (19).** Compounds scoring above 18 of 20, i.e. severe signs in nearly all categories, up to the maximum score of 20.
- **Graded across the range.** Severity grades 0/1/2/3 = 56/87/40/57.

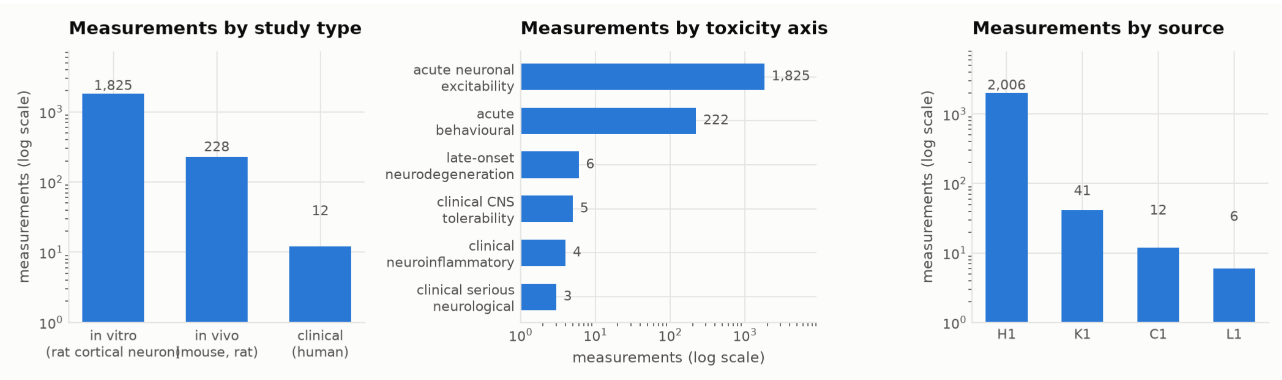

Figure 1. *What the dataset contains. Log scales: the in vitro arm dominates by count, but the in vivo and clinical arms carry the severity grades. Source H1 is Hagedorn 2022, K1 Miller 2024, L1 Kuroda 2025, C1 the FDA labels.*

2 Main findings and conclusions

Finding 1 — sequence predicts in vivo CNS tolerability better than the in vitro assay does

Across the 181 oligonucleotides that carry both readouts, the measured in vitro calcium-oscillation score separates tolerable from intolerable compounds with an AUC of 0.73, while a score computed from the sequence alone reaches 0.89. On the held-out set — 19 oligonucleotides against a different target gene, held out by the original authors — the sequence model reaches 89.5% accuracy.

The explanation is measurement noise, and the dataset contains the evidence for it. One control oligonucleotide appears 14 times in the source table, having been run on 14 independent plates. Its scores range across 17.3% coefficient of variation (mean 92.4, SD 16.0). A single in vitro measurement carries that noise; a sequence-derived score does not. **The practical consequence for anyone building a model: do not treat the in vitro assay as ground truth to be predicted, and do not discard a compound on one plate run.**

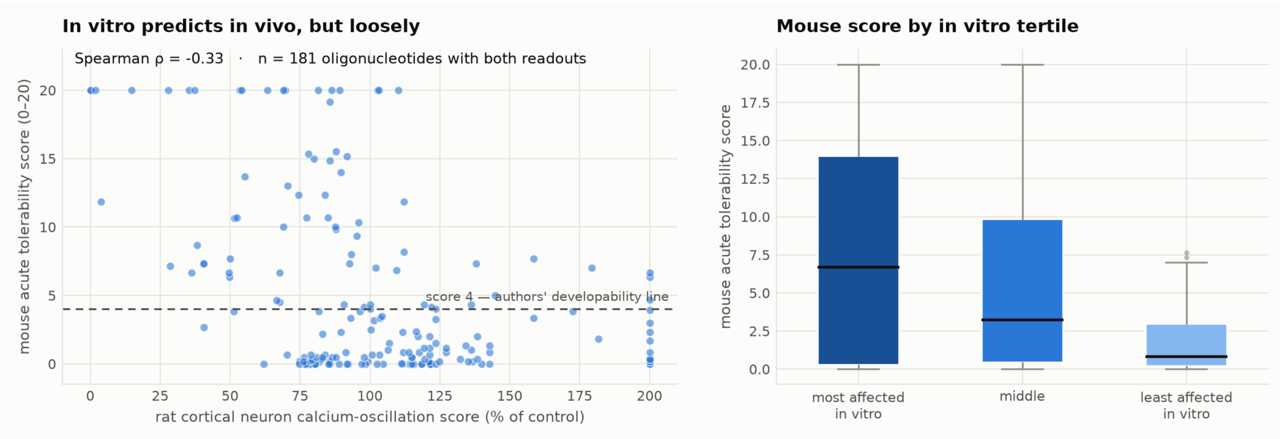

__Figure 2.__ In vitro to in vivo translation, n = 181. The relationship is real and monotone across tertiles (median mouse score 6.67, 3.25, 0.83) but loose at the level of an individual compound (Spearman ρ = –0.33).

Finding 2 — guanine content is a graded risk factor, not a threshold

Median mouse tolerability score rises monotonically with the number of guanines in the oligonucleotide: 0 G → 0.3 (n=81); 1 G → 3.2 (n=37); 2 G → 7.3 (n=16); 3 G → 7.7 (n=30); 4 G → 15 (n=11); 5 G → 20 (n=3). Position matters more than count. Binning by the length of the guanine-free stretch measured from the 3′ end gives median scores 0-4 nt → 12.3 (n=45), 5-9 nt → 6.8 (n=28), 10-14 nt → 2.5 (n=21), 15-20 nt → 0.3 (n=87). A guanine close to the 3′ end is the single strongest sequence warning in this dataset.

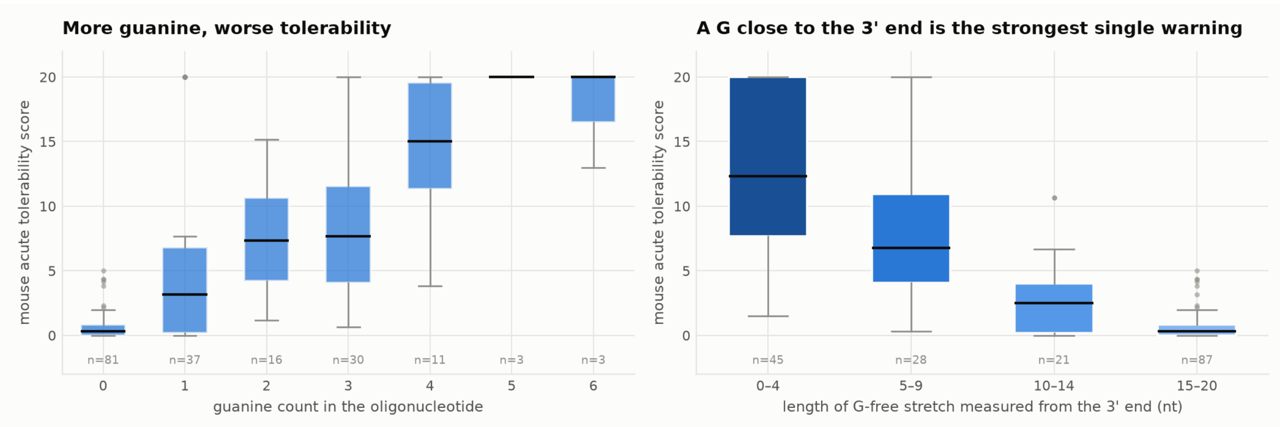

__Figure 3.__ Both guanine relationships, with group sizes shown. The extreme guanine-count bins are small (n = 3), which is why the count relationship should be read as a trend and the 3′-position relationship, with 21–87 per bin, as the more reliable of the two.

Finding 3 — formulation can override sequence and chemistry entirely

At a fixed dose of the same molecule, adding calcium to the injectate reduces the acute tolerability score from 19.5 to 1 of 20 as calcium rises from 0 to 32 mM. That is close to an on/off switch, produced without changing a single nucleotide. Magnesium produces a similar but shallower effect.

This has a direct implication for how the dataset should be modelled. A model trained on sequence and chemistry alone, on data pooled across studies that used different vehicles, will attribute a formulation effect to the molecule. The dataset therefore carries `formulation_ca_mM` and `formulation_mg_mM` as first-class columns rather than burying them in a methods note.

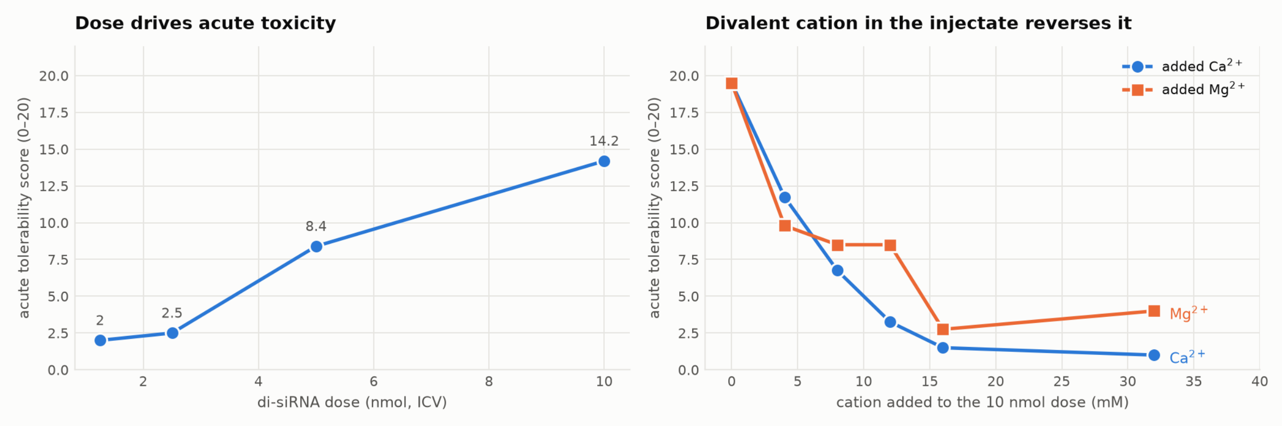

Figure 4. *Dose-response and cation rescue, both from source K1, both on the same 0–20 scale as the core dataset.*

Finding 4 — two published sources disagree, and the dataset makes the disagreement visible

Source K1 shows divalent cations abolishing acute CNS toxicity. Source O1 reports that divalent cation supplementation at 1–100 mM did *not* alter the acute response it measured. Both can be correct, because they measure different phenotypes: K1 scores an *activation* syndrome (hyperactivity, tremor, seizure) and O1 scores an *inhibition* syndrome (progressive loss of motor function, scored 3 h after dosing on a 0–7 scale). A model trained on a pooled “CNS toxicity” label would be learning across a real mechanistic boundary. This is why every measurement carries a `tox_axis`, and why `docs/SCORING_INSTRUMENTS.md` states explicitly which axes may be pooled.

Finding 5 — acute and late-onset toxicity are separable, and separably recorded

Source L1 reports gapmers that are *not* acutely toxic — one produces acute signs that resolve within a day, three produce none — yet cause hypoactivity and motor loss appearing three or more days after dosing, severe enough that two required humane sacrifice at day 7 and one of four rats given the compound intrathecally died at day 14. Acute screening cannot see this. The dataset keeps the two windows on separate axes so that a model is not trained to call a late-onset toxin safe.

Conclusions

- Sequence and modification position, taken together, carry enough signal to predict acute CNS tolerability at clinically useful accuracy — 89.5% on a genuinely held-out target.
- The commonly used in vitro screen is noisier than the sequence model it is meant to validate; its coefficient of variation is measurable from this dataset at 17.3%.
- Formulation is a confounder of the first order and must be modelled, not assumed constant.
- “CNS toxicity” is at least four distinct phenomena. Collapsing them into one label discards mechanism and creates contradictions that are artefacts of the labelling, not the biology.

3 How the data were produced

This is a **curation** dataset: no new wet-lab experiment was run. Its contribution is the structuring, verification and harmonisation of experimental results that exist in the public record but not in any usable form. What follows is a summary; the methodology document gives the full protocol.

Source selection

Sources were required to report a CNS-specific toxicity outcome for an identified oligonucleotide, and were prioritised by whether they also published the sequence. A source that reports toxicity with sequences is worth far more than one that reports toxicity alone, because only the former can train a model. Sources reached by web-search summary alone were not used for any numeric value.

ID	Source	Contribution	Licence	Oligos	Measurements
C1	US FDA / Biogen Inc. 2024/2026 FDA prescribing information	regulatory primary	US Government work / public domain	2	12
H1	Hagedorn PH 2022 Nucleic Acid Therapeutics	primary supplementary data	CC BY 4.0	1825	2006
K1	Miller BR 2024 Molecular Therapy Nucleic Acids	primary supplementary data	CC BY-NC	7	41
O1	O'Rourke JJ 2026 Nucleic Acids Research	primary fulltext instruments only	CC BY-NC	0	0
L1	Kuroda T 2025 Molecular Therapy Nucleic Acids	primary supplementary data	CC BY-NC	5	6

Table 1. *The source registry, reproduced from* `data/sources.csv`*. O1 contributes measurement instruments and a contradictory finding, but no rows.*

Acquisition

Supplementary files were retrieved through the Europe PMC REST endpoint after the PubMed Central interface proved to gate binary downloads behind a JavaScript proof-of-work challenge. Regulatory labels were read from DailyMed. Every retrieved file is committed alongside the code, so the dataset remains rebuildable if a publisher URL rots.

Computational processing

- **Sequence and modification position.** In source H1 the printed sequence encodes chemistry in case. This was not taken on trust: the count of upper-case characters was checked against the paper's own declared LNA count for all 1,825 rows, with zero mismatches, before position was read out of case.
- **Typeface parsing.** In source L1 the chemistry is encoded in **bold** (LNA) and ***bold-italic*** (2′-MOE), which plain text extraction destroys. Per-position chemistry was recovered by reading the PDF's span styling, giving 3-10-3 and 4-12-4 LNA gapmers and one 5-10-5 2′-MOE gapmer.
- **Table parsing.** Source K1's tolerability table was parsed from the supplementary PDF by a state machine over the text layer, recovering 41 dosing groups with their injectate cation concentrations.
- **Derived fields.** Base composition, G+C content, longest guanine run, 3′ guanine-free stretch and gap geometry are computed from the printed sequence and marked as derived. Nothing else is computed.

As an end-to-end check that the transcription is faithful, the predictive model published by the source authors was re-implemented from their supplementary methods and run over the restructured table: it reproduces their own published score column for all 1,825 rows, with zero mismatches beyond rounding.

4 Indicators and predictor variables

How the toxicity indicators were measured

Indicator	Instrument	Range / units	n
Acute tolerability score	Modified functional observational battery (Oligonucleotide Safety Working Group), five categories each 0–4, summed; scored over 1 h after a single ICV bolus; 4–6 mice per group, averaged	0–20	222
Calcium-oscillation score	Spontaneous calcium oscillations in rat E19 primary cortical neurons, fluo-4 AM on FLIPR; 1 point per 1-s read exceeding 50% of mean control amplitude over a 300 s read	% of control (lower = greater effect)	1,825
Late-onset tolerability	Five-category 0–4 scale applied over days 1–21; open-field locomotion and body weight alongside	0–20 (published as figures only)	6
Clinical adverse-event incidence	Randomised controlled trial safety reporting, as printed in FDA prescribing information	% of arm	12

Table 2. *Each instrument is reproduced verbatim from its source in* `docs/SCORING_INSTRUMENTS.md`*, including which instruments may legitimately be pooled.*

Distribution of the predictor variables amongst tested oligonucleotides

Lengths run 14–25 nucleotides, with a strong mode at 16 and 20. 1,731 compounds have a determined DNA gap length; 99 are mixmers whose modified residues are not contiguous, for which a single gap length is undefined and is left empty rather than guessed. Guanine count is strongly right-skewed (median 1), which is by design: the source library was built to interrogate guanine as a risk factor and deliberately over-samples guanine-poor sequences.

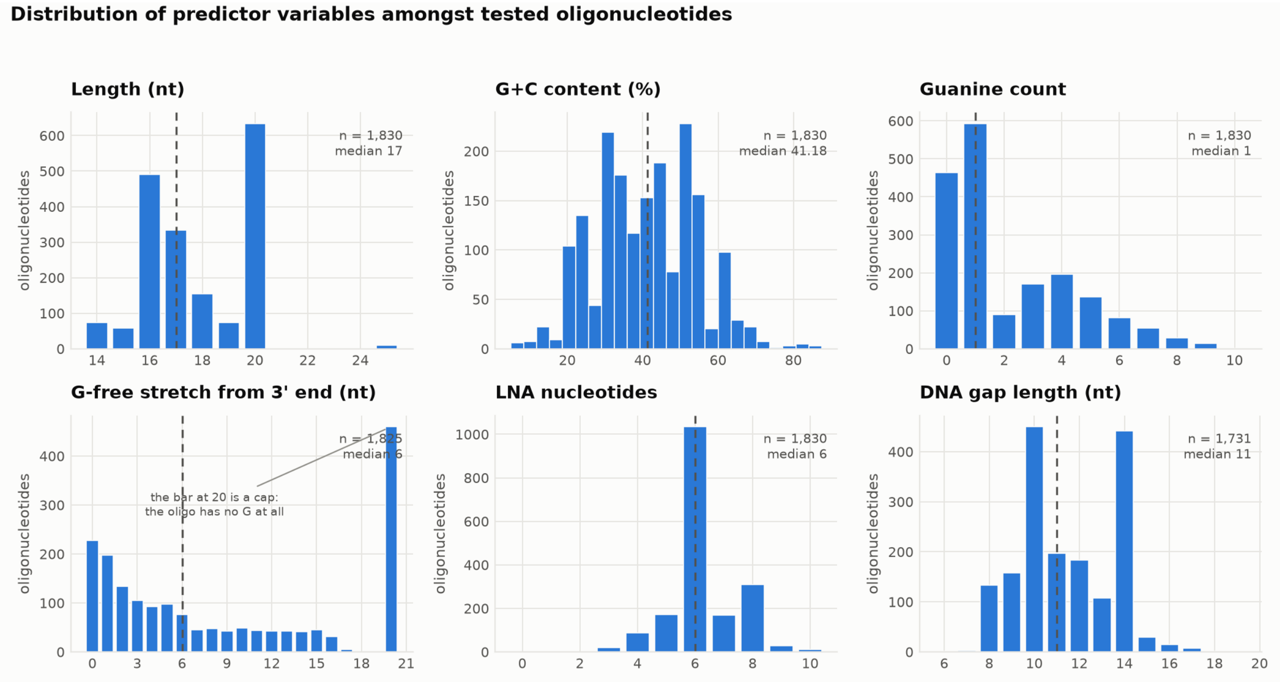

Figure 5. *The six principal predictors. The spike at 20 in the 3′ guanine-free panel is a cap defined by the source, meaning the oligonucleotide contains no guanine anywhere; it is not a measured length of 20.*

Distribution of the toxicity outcome

240 of 2,065 measurements carry an ordinal severity grade. The remaining 1,825 are the in vitro calcium-oscillation readings, deliberately left ungraded: they are a continuous measure of a different quantity and the source defines no severity bands for them, so grading them would mean inventing thresholds.

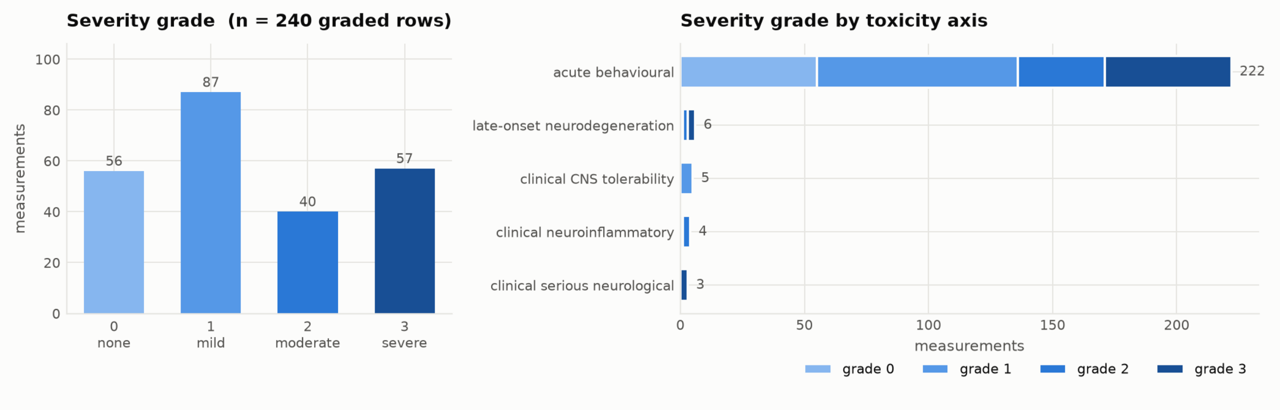

__Figure 6.__ Severity distribution. All four grades are well represented (56/87/40/57), which matters: a dataset of positives alone cannot train a classifier. The grade boundary between 1 and 2 reproduces the source authors' own developability line — 112 of 181 in vivo rows (61.9%) fall at grade ≤ 1, against their stated "roughly 60%".

5 The gap in public data that this addresses

Oligonucleotides delivered into cerebrospinal fluid have a specific, well-documented toxicity problem, and two approved drugs carry CNS warnings on their labels. Yet the public data needed to predict that toxicity from a candidate's design has been effectively unavailable, for three separate reasons that this dataset addresses.

- **The data existed but was not in usable form.** The largest public sequence-resolved CNS dataset sits inside a single supplementary spreadsheet, behind a download challenge, with chemistry encoded in text case and no position table, no controlled vocabulary, and no join to any other source. It is generally cited for 148 compounds; it holds 1,825.
- **Modification position was not machine-readable anywhere.** The challenge asks for the location of every chemical modification. Sources supply this as a case convention, as typeface, or as a motif string, and no public CNS dataset had normalised it. This release provides 32,569 position-level records with an explicit provenance basis on each.
- **Findings were scattered across incompatible instruments.** At least five distinct scoring scales are in use. Placing them in one schema, with the instruments documented and the incompatible ones explicitly marked as un-poolable, is a prerequisite for cross-study modelling that no previous public resource provided.

The dataset also makes visible two things that only appear when sources are placed side by side: the measured noise of the standard in vitro screen, and a direct contradiction between two 2024–2026 papers on whether divalent cations mitigate CNS toxicity. Neither is visible from within any single publication.

6 Using this data to build a predictive model

The claim that this dataset supports predictive modelling is demonstrated rather than asserted. `src/baseline_model.py` reads only the released CSVs and trains against the source's own train/test split, so the held-out set is the one the original authors held out: 19 oligonucleotides targeting a *different gene* from the compounds used for fitting. That is a generalisation test, not a random split.

Model	Inputs	Held-out AUC	Held-out accuracy
Published linear model	base composition + 3′ guanine-free length	0.929	89.5%
Measured in vitro assay	calcium-oscillation score	0.929	—
Logistic regression fitted here	12 sequence and geometry features	0.857	78.9%

Table 3. *Predicting the source authors' own developability line (tolerability score > 4) on 19 held-out compounds, 7 of them positive. The fitted logistic regression uses 138 training compounds — the subset of the 181 in vivo rows for which all twelve features are defined; the remaining 24 are mixmers, which have no single DNA gap length. The two sequence-only rows use all 181.*

Two results are worth stating plainly. First, the published five-parameter model reproduces at 89.5% accuracy on the held-out set, which independently confirms that the restructured table is faithful. Second, **a richer twelve-feature logistic regression fitted here did not beat it** (AUC 0.86 against 0.93). With 138 labelled training compounds, added features cost more in variance than they buy in signal. That is a useful negative result for anyone planning to model this data: the constraint is labelled in vivo examples, not features.

A recommended build order

- **Reproduce the baseline first.** If your pipeline cannot reach roughly 89% on the held-out set with base counts and the 3′ guanine-free length, the pipeline is wrong, not the data.
- **Model the two readouts jointly, not sequentially.** The in vitro score is available for all 1,825 compounds and the in vivo score for only 181. Treating the in vitro score as a noisy auxiliary task, with its measured 17.3% coefficient of variation as a known noise floor, uses ten times more data than in vivo-only training.
- **Use the position table, not just base counts.** The published model uses composition only. Position-resolved chemistry is now available for 1,830 compounds and is entirely unexploited.
- **Include formulation.** Calcium and magnesium concentration are dataset columns. A model that ignores them will attribute a vehicle effect to a sequence.
- **Respect the axes.** Train separate heads for acute and late-onset toxicity, or at minimum condition on `tox_axis`.

What would most improve the next version

More labelled in vivo examples, and chemistry breadth. The core is one chemistry class — LNA/DNA full-phosphorothioate gapmers — which is a strength for isolating sequence effects and a limitation for generalising. Two published datasets documented here as instruments only (non-human-primate and 2′-MOE) would address both, and are the top priority for v1.1.

7 Limitations, stated plainly

Per-compound purity is absent from the literature. `purity_pct` is `NOT_REPORTED` for all 1,839 oligonucleotides. Journals essentially never print per-compound purity or observed mass alongside toxicity results. What is captured, where the source states it, is the purification and identity-confirmation *method* — present for 1,825 of 1,839. No purity value has been estimated. This is the largest gap between this dataset and the challenge's description of one, and it is a property of the published record rather than of the curation.

- **The in vitro arm is rat, not human.** Only 12 of 2,065 measurements are human-derived, and all of them are clinical. We searched specifically for human iPSC-derived neuron or organoid data on oligonucleotide CNS toxicity and found reviews describing it as promising but no published, sequence-resolved source. The field's standard predictive screen is a rodent primary-neuron assay. This is the most important scientific gap the dataset reveals.
- **Chemistry is narrow.** 1,825 of 1,839 compounds are LNA/DNA full-phosphorothioate oligonucleotides from a single study — 1,726 conventional gapmers and 99 mixmers.
- **Grades are provisional.** Every grade carries the rule that produced it, but none has been reviewed by a subject-matter expert; all ship as `provisional`.
- **One row is one group, not one animal.** In vivo scores are group means over 4–6 mice. Within-group variance is not recoverable from the published data.
- **Publication bias.** The 40% failure rate in the core library is a property of a library built to interrogate toxicity, not a base rate for CNS oligonucleotides generally.

Quality control

`qc/validate_dataset.py` runs 30 structural and provenance checks — key uniqueness, referential integrity, controlled-vocabulary conformance, grade range, sequence self-consistency, modification-table completeness and contiguity, and the rule that **no numeric readout may exist without a named source table or figure**. All 30 pass, and the script exits non-zero on any failure so it can gate a release.

Access and licence

CC BY 4.0 for everything created here. Row-level content carries its source's terms in a `redistribution` column: 2,018 of 2,065 measurements (97.7%) are CC BY 4.0 or US public domain and are reusable for any purpose including commercially; the remainder derive from CC BY-NC sources, are individually marked, and are removable with a one-line filter. No registration, no data-use agreement, no embargo. Full plan in `docs/PADP.md`.

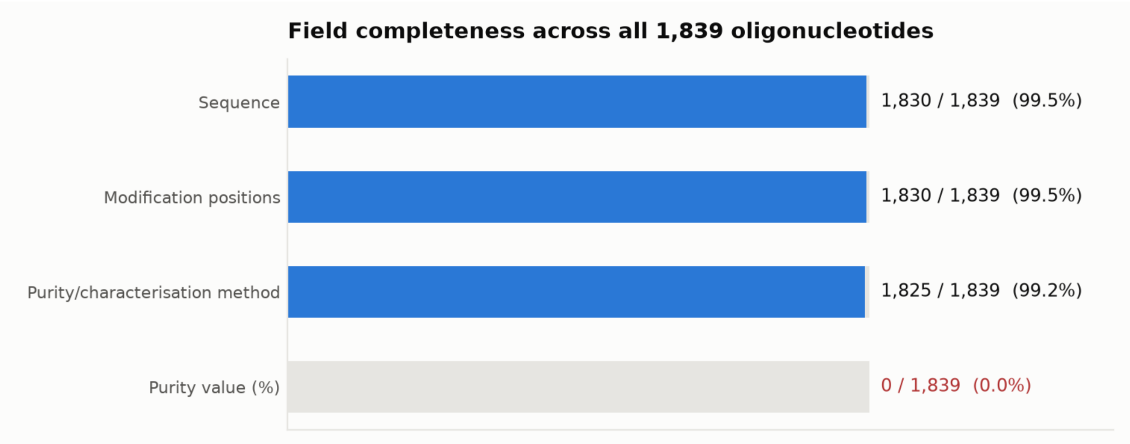

Figure 7. *Field completeness, including the purity gap stated above. Nothing has been filled in to improve this picture.*